

Multiple Choice (10 marks)

1. Consider the Bayesian network given below. How many independent parameters are needed for this Bayesian Network $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 2
 - (b) 4
 - (c) 8
 - (d) 16
2. Predicting the amount of rainfall in a region based on various cues is a _____ problem.
 - (a) Supervised learning
 - (b) Unsupervised learning
 - (c) Clustering
 - (d) None of the above
3. For polynomial regression, which one of these structural assumptions is the one that most affects the trade-off between underfitting and overfitting:
 - (a) The polynomial degree
 - (b) Whether we learn the weights by matrix inversion or gradient descent
 - (c) The assumed variance of the Gaussian noise
 - (d) The use of a constant-term unit input
4. As the number of training examples goes to infinity, your model trained on that data will have:
 - (a) Lower variance
 - (b) Higher variance
 - (c) Same variance
 - (d) None of the above
5. Which among the following prevents overfitting when we perform bagging?
 - (a) The use of sampling with replacement as the sampling technique
 - (b) The use of weak classifiers
 - (c) The use of classification algorithms which are not prone to overfitting
 - (d) The practice of validation performed on every classifier trained

True / False (10 marks)

Answer True or False

6. True or False: Grid search runs reasonably slow for multiple linear regression due to its exhaustive nature in hyperparameter tuning.
7. True or False: Principal Component Analysis (PCA) is an unsupervised learning technique used for dimensionality reduction.
8. True or False: As of 2020, several models achieve over 98% accuracy on the CIFAR-10 dataset.
9. True or False: The models obtained from best-subset selection and forward stepwise selection will always yield the same results in linear regression.
10. True or False: Residual connections are integral components of both ResNets and Transformers, enhancing their training and performance.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the modifications needed in the EM algorithm to transition from finding maximum likelihood estimates to maximum a posteriori estimates, focusing on the maximization step.
12. Discuss the implications of applying regularization in linear regression, focusing on how certain penalties can lead to zeroed-out coefficients and their impact on model performance.

End of Paper